`rewrite`: a programming language that learns its mathematics from rule files

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Abstract

`rewrite` is a small interpreted language for symbolic and numerical mathematics, written in the early 1990s and revived in 2026. Unlike most computer algebra systems, its kernel contains no mathematics: data types, operators, the laws of algebra, differentiation and integration rules are all defined in plain-text include files and applied by term rewriting. Rule sets ("algebras") are first-class objects that can be combined into sequences and fixpoint iterations. A `LIFT` statement transfers all rules of one set of operations to another, so that, for example, the rules of an abelian group written for $+$ also apply to $\cdot$ in a field. Numerical evaluation is interleaved with rewriting: as soon as all arguments of a function are numbers, an attached C function computes the value. We describe the design of the language, its revival on current 64-bit compilers, and its extension by static subsorts, which add exact integers and fractions as sorts below the real numbers without changing any rule of the library. As a case study, the rule-based integrator Rubi was translated into `rewrite`. A translator converts 90% of Rubi's 7422 rules automatically; for this, the language gained ordered rule lists with a backtracking matcher for optional pattern parts, commutative operators and condition lists, symbolic helper functions and special functions. On 1540 integration problems from four independent textbook collections, 55.0% of the results are verified correct by numerical differentiation.

1 Introduction

Computer algebra systems such as Mathematica [17] or SymPy [10] come with a large body of built-in mathematical knowledge. A user can add rules, but the core simplification and the calculus routines are part of the system and are mostly hidden from the user. Term rewriting systems such as Maude [5] take the opposite view: they provide a general rewriting engine and leave the mathematics to the user, but they are usually not aimed at numerical work.

`rewrite` sits between these positions. Its interpreter knows how to parse expressions with user-defined operators, how to match and replace terms, and how to call C functions for numerical values. Everything else is defined in include files: the sort `real`, the operators `+`, `*` and `^`, the group, ring and field laws, the differentiation rules, and numerical algorithms such as Simpson integration. The language was designed by the author between 1992 and 1996 on Unix and the Amiga, on top of a scanner and parser framework by Axel T. Schreiner. In 2026 it was ported to current compilers and extended with a test suite, exact numbers and a rule-based integration library.

The main contributions of this paper are:

- a description of a language in which mathematics is a library of rewrite rules, including the `LIFT` mechanism for transferring rules between algebraic structures;
- the interleaving of symbolic rewriting and numerical evaluation, which allows numerical algorithms to be written as rewrite rules;

- static subsorts with value conversion, which add exact integers and fractions to the language as a library, with the rules of the real numbers inherited through `LIFT`;
- an automatic translation of the rule-based integrator Rubi [11] into `rewrite`, with a measurement on independent test problems and a discussion of what the translation requires from the host system.

2 Language design

2.1 Sorts, operators and numbers

A program first declares sorts (types) and operators. The following complete program defines real numbers with three operators and evaluates expressions:

```
SORT real                                   # a data type
NUM real                                    # numbers typed in have sort real
FUNC + : real,real >> real LA2 ATT plus     # left-associative, priority 2
FUNC * : real,real >> real LA3 ATT mal      # binds tighter than +
FUNC ^ : real,real >> real RA  ATT hoch     # right-associative
1+2*3                                       # prints 7
2^3^2                                       # prints 512
CONST x,y : real
x+y*2                                       # prints x + y * 2
```

The format words `LA1`–`LA5`, `RA` and `RU` define left-associative binary operators of five priorities, right-associative operators and right unary operators such as the factorial. `ATT name` attaches a C function from a table of user functions. It is called whenever all arguments of the function are numbers; otherwise the term stays symbolic. `OTHER` adds further meanings to an existing name for other argument sorts, so that `+` can mean addition of reals, complex numbers or vectors. `BRA` defines functions written with brackets, such as a scalar product $\langle a, b \rangle$.

Each sort has its own representation in C. Real numbers are `double` values; the standard library defines complex numbers, real vectors and matrices, whose values are C arrays. Operations on whole vectors and matrices are executed by C functions, so they run at compiled speed.

2.2 Rules and algebras

Rules are collected in named rule sets, called *algebras*. A rule has a left side, a right side, an optional condition and an optional block of instructions that is executed before the replacement:

```
VAR A : real
ALG simp
SPEC    simp : A*1 = A              # SPEC starts a new rule set
ADDSPEC simp : A+0 = A              # ADDSPEC adds a rule
CONST u : real
REWR (u*1)+0 : simp                 # prints u
```

`VAR` declares meta variables that match any term of their sort. `NUM` declares meta variables that only match numbers, which makes rules like "evaluate if the argument is a number" easy to write. Conditions test whether a term occurs in another one (`IN`, `NIN`), compare terms in a total order (`LB`, used for sorting sums), or evaluate any expression with a boolean result.

Algebras are combined into lists. A list in parentheses is applied repeatedly until no rule changes the term any more. The standard rule list of the library is defined like this:

```
SPEC All : Std,Doit,(Linear,Compl,diff,GrRi),AGroup,integr,
```

```
        (Subst,GrRi),Std,Zus,unaerweg
```

With the library loaded, `REWR D(sin cos x,x) :  All` returns `- cos (cos (x ))* sin (x )`.

2.3 Transferring rules with LIFT

A distinctive feature of `rewrite` is the `LIFT` statement. It declares that all rules valid for one set of functions and constants also hold for another set. The standard library writes the laws of a group once and transfers them step by step:

```
LIFT (Ag+Ag,Ag-Ag,-Ag,0g) TO (Aag+Aag,Aag-Aag,-Aag,0ag)
LIFT (Aag+Aag,Aag-Aag,-Aag,0ag) TO (Ar+Ar,Ar-Ar,-Ar,0r)
LIFT (Aag+Aag,Aag-Aag,-Aag,0ag) TO (Ar*Ar,Ar/Ar,mi(Ar),1r)
```

The last line reuses the rules of an abelian group for multiplication, division, the multiplicative inverse `mi` and the neutral element 1 of a ring. In the same way, linearity is written once for a generic linear function and lifted to differentiation, integration, Taylor expansion and summation. Each symbol can be lifted from one other symbol, so the lifted operators form a tree, and a rule applies to a term whose operator reaches the operator of the rule along this tree. This corresponds closely to the algebraic hierarchies of modern proof assistants [15], but here it acts directly on the rewrite rules.

2.4 Exact numbers as subsorts

Exact arithmetic was added in 2026 without changing the rules of the library, by two sorts and two new statements. The sorts `integer` (arbitrary size) and `rational` (fractions) are implemented in C with the GMP library [7] and declared in an include file:

```
SORT integer
SORT rational
ATT integer  show_int,cmp_int,cpy_int,del_int
ATT rational show_rat,cmp_rat,cpy_rat,del_rat
SUBSORT integer,rational ATT int2rat     # every integer is a fraction
SUBSORT rational,real ATT rat2real       # every fraction is a real number
OTHER + : rational,rational >> rational ATT rat_plus
OTHER / : integer,integer >> rational  ATT int_div
...
LIFT (A+A,A-A,A*A,A/A,-A,mi(A),A^A) TO (Aq+Aq,Aq-Aq,Aq*Aq,Aq/Aq,-Aq,mi(Aq),Aq^Az)
INTNUM integer ATT read_int              # 12 is an integer, 0.5 stays real
```

`SUBSORT` declares that every value of the first sort is also a value of the second, with a C function that converts the value. Sorts remain static: the sort of a term follows from its parts when it is built, and a value never changes its sort at run time. A function declared for a wider sort accepts arguments of its subsorts; if several variants of an operator fit, the one with the fewest conversions is chosen, and the values are converted only when the function is calculated. Thus `1/3+1/6` uses the rational `+` and prints `(1/2)`, `2^100` is exact, and `1/3+0.5` uses the real `+` and prints `0.833333`. The `LIFT` line lets `rational` inherit all rules of `real`, and a second one does the same for `integer`; `x*1+0*x+(1/2)*x` simplifies to `(3/2) * x` with the unchanged rules for real numbers.

One change in the interpreter made this work. When a rule written for `real` rewrites a term with exact numbers, the operators of the new term are chosen again for the sorts of their actual arguments, from the leaves upwards and after each numerical step. The original `LIFT` mechanism replaced all operators of one sort by those of another; that was correct for homogeneous structures like groups and rings, but with subsorts the arguments of one term

have different sorts. Re-choosing the variant repairs this and makes rules for `real` produce exact results whenever their inputs are exact. The declarations are a classic instance of order-sorted algebra [6]; the novelty here is that the subsort relation, the conversions and the inheritance of rules are all defined in the library.

2.5 Interleaving rewriting and numerics

After every single rewrite step, the interpreter evaluates all subterms whose arguments are numbers. Constants that have a value are replaced by their value when a term is copied. Together with the procedural part of the language (`FOR`, `WHILE`, `IF`, blocks with local constants, and user functions), this allows numerical algorithms to be written as rewrite rules. The library defines Simpson integration as a rule whose `DO` block computes the value:

```
ALG rechne
SPEC rechne : I(N,M,A,_x)=_myerg DO { _myerg=0;_zw1=0;_zw2=0  # Simpson
        FOR n=N+_step/2 TO M-_step STEP _step DO { _x=n;_zw1=_zw1+A
        _x=n+_step/2
        _zw2=_zw2+A }
        _x=M-_step/2
        _zw1=_zw1+A
        _x=N;_myerg=A+4*_zw1+2*_zw2
        _x=M;_myerg=(_myerg+A)*_step/2/3
        _x=dummi }
```

A second rule set first substitutes the integration variable by the internal variable `_x`. With `_step=0.01`, `REWR I(0,pi,sin(x),x) : numerik,rechne` returns 2. The same mechanism gives Newton's method and the search for minima and maxima. The symbolic part can prepare the numerical part: `newton` expects the derivative as an argument, which is computed symbolically by `D`.

The original application of the language was physics. The library contains physical constants, units and formulas, and an example program computes Bragg angles, structure factors and X-ray absorption of lithium niobate crystals from tabulated atomic data.

3 Case study: rule-based integration

Differentiation is easy to express with rewrite rules, because every rule takes the term apart and the product and chain rules always apply. Integration is harder. Complete decision procedures exist for elementary functions [12, 2], but they are not pattern based. Rubi [11, 13] showed that an extensive system of carefully ordered rules with conditions can integrate a very large class of functions. Its 199 rule files contain 7422 rules in Mathematica syntax, published under the MIT license. We first translated a small closed subset by hand and then wrote a translator for all of them, to test how far the mechanisms of `rewrite` carry.

3.1 Pattern features

Rubi's patterns rely on features of the Mathematica pattern matcher that the original `rewrite` did not have. A typical rule is

```
Int[(a_. + b_.*x_)^m_, x_Symbol] := (a + b*x)^(m + 1)/(b*(m + 1)) /;
    FreeQ[{a, b, m}, x] && NeQ[m, -1]
```

where `a_.` matches a missing summand with default 0 and `b_.` a missing factor with default 1. The pattern also matches $x\,b + a$, because Mathematica matches sums and products in any order, and the condition combines several tests. We added three corresponding features to the

language: `OPT a,b` declares meta variables that may be missing, `DEFAULT + = 0`, `DEFAULT * = 1` and `DEFAULT RIGHT ^ = 1` give the values of missing operands, `COMM +,*` declares commutative operators, and a rule can have a list of conditions, all of which must hold. `NIN(x,a,b,m)` tests that x occurs in none of a, b, m, which is Rubi's `FreeQ`.

A first implementation expanded every rule into all its variants when it was defined, so that the matcher stayed unchanged. The number of variants grows exponentially with the number of optional parameters and commutative products, however: the median Rubi rule would have about 1900 variants. Rules defined while one of these features is active are therefore kept in an ordered *rule list*, as in Mathematica, and matched by a separate backtracking matcher. Bindings of meta variables are recorded on a trail, so a failed alternative undoes exactly its own bindings. Swapped operands and missing optional operands are tried in turn, and the conditions are checked as part of the match, so a failing condition makes the matcher try the next alternative and then the next rule. A precomputed set of possible head symbols per rule skips most rules of a large list without running the matcher. With these features the rule above translates almost one to one:

```
ADDSPEC RubiRules : IF NIN(Xv,O_a,O_b), (neq(P_m,(-1)))
        Int(((O_a)+((O_b)*(Xv)))^(P_m),Xv) = ...
```

Here `P_m` is a required and `O_a`, `O_b` are optional meta variables. One rule covers $(a+bx)^m$, $(bx+a)^m$, $(a+x)^m$, $(bx)^m$ and x^m, with numeric or symbolic coefficients.

3.2 Automatic translation

The translator (900 lines of Python) parses Rubi's rule files with a small Pratt parser into Mathematica's full form and writes one `rewrite` file per Rubi file, in Rubi's loading order and with the original of every rule as a comment. Pattern variables become meta variables, negative numbers in patterns become `NUM` variables with a condition, local definitions (`With`, and `Module` with values only) are substituted, helper predicates that Rubi defines inside its rule files are inlined, and a predicate applied to a list, like `LinearQ[{u,v},x]`, becomes a conjunction. e^u is matched as a power with base e, so that the rules for F^u apply to it, as in Mathematica.

Helper and special functions. Rubi's conditions and right-hand sides use about a hundred helper functions, such as `ExpandIntegrand`, `PolyQ`, `BinomialQ`, `Subst` or `ExpandTrigReduce`. They cannot be ordinary `rewrite` functions, which are only evaluated when all arguments are numbers. We added symbolic C functions instead: they receive the argument terms and return a new term, or nothing if they do not apply. About 60 of them are implemented, from predicates on the form of a term to polynomial division, the product-to-sum reduction of trigonometric powers, and Rubi's measures of simplicity based on the size of a term. Their results are exact when their inputs are. Many results of Rubi need special functions; 27 of them, from the error function and the Fresnel and exponential integrals to the incomplete Gamma function, elliptic integrals in Carlson's form [3] and the hypergeometric function ${}_2F_1$, were added as C functions with Mathematica's conventions, each with a derivative rule. Their values agree with mpmath [9] to within 10^{-11}.

The host system's simplification. Rubi relies on Mathematica's automatic simplification, which `rewrite` does not have. Most of the work after the first translation went into reproducing the parts Rubi depends on. A normal-form rule set writes $x-y$ as $x+(-1)\,y$, x/y as $x\,y^{-1}$ and $\sqrt{x}$ as $x^{1/2}$, but, unlike our first version and like Mathematica, it does not multiply out products with sums: Rubi's rules for $(c+d\tan x)/(a+b\tan x)$ expect the product. Polynomials in the integrand are brought into the form $c_0+c_1x+c_2x^2+\ldots$ with collected coefficients and their numeric content in front, and every rule result is simplified trivially (u^0, u^1, $1\cdot u$, $0+u$). In

Mathematica, some rules that make no progress are harmless because exact arithmetic or other rules prevent a loop; in `rewrite` they looped. The rule list therefore skips a rule whose result equals the term, contains a NaN or a division by zero, or equals one of the last terms the list replaced (a cycle such as expanding and factoring again), and a rule that keeps rewriting the same term. The matcher also had to learn that an optional variable already bound to its default value may be missing a second time, as in $(c + d\tan(e + fx))/(a + b\tan(e + fx))$ with $e = 0$.

The complete integration is the algebra list

```
SPEC RubiIntegrate : (RubiNorm,IntRubi,RubiRules),RubiCollect,IntOut,All
```

where `RubiCollect` combines like summands of the result and `IntOut` restores the usual output form.

3.3 Results

The translator converts 6713 of Rubi's 7422 rules (90%). The largest group of the remaining rules needs complex numbers (245 rules); others use patterns with function variables (`F_[x]`), Rubi's substitution machinery (`SubstFor`, `FunctionOfQ`, `DerivativeDivides`), general pattern tests (`MatchQ`) or the generalized hypergeometric function. Loading all translated rules takes about half a second, and a typical integral a fraction of a second. Table 1 shows examples.

Integrand	Result of `REWR Int(f,x) : RubiIntegrate`
`1/(1+x^3)`	`(1/3)*ln(1+x) + 0.57735*arctan(-0.57735+1.1547*x) - (1/6)*ln(1+x^2-x)`
`x/(1+x^4)`	`(1/2)*arctan(x^2)`
`sqrt(1-x^2)`	`(1/2)*arcsin(x) + (1/2)*(1-x^2)^(1/2)*x`
`x*sqrt(1+x)`	`(2/5)*(1+x)^(5/2) - (2/3)*(1+x)^(3/2)`
`sin(x)^2*cos(x)^3`	`(1/3)*sin(x)^3 - (1/5)*sin(x)^5`
`1/(1+tan(x))^2`	`(1/2)*ln(cos(x)+sin(x)) - (1/2)/(1+tan(x))`
`x*arcsec(x)`	`(1/2)*x^2*arcsec(x) - (1/2)*(1-1/x^2)^(1/2)*x`
`exp(-x^2)`	`0.886227*erf(x)`
`sin(x)/x`	`sinint(x)`
`1/sqrt(1-x^4)`	`ellipticf(arcsin(x),-1)`
`(1+x^3)^(1/3)`	`hyp2f1((-1/3),(1/3),(4/3),-x^3)*x`

Table 1: Indefinite integrals computed with the automatically translated Rubi rules (output slightly condensed). Rational coefficients are exact; irrational ones such as $1/\sqrt{3}$ and $\sqrt{\pi}/2$ are real numbers.

To measure the translation, a test harness runs problems of Rubi's own test suite [14] through `RubiIntegrate`. It does not compare the result with Rubi's antiderivative, which may differ by a constant or in form, but checks it numerically: the derivative of the result, computed by a five-point difference quotient, must equal the integrand at several points where the integrand is real, with fixed values for symbolic parameters. The four collections of independent problems in the test suite (from the textbooks of Apostol, Hearn, Stewart and Timofeev, 1540 problems, of which 1539 use only functions that `rewrite` has) gave the results in Table 2.

The development of the translation went through several such measurements; each round of fixes, such as the normal form without multiplying out, the matching of optional variables bound to their default, the special functions and the missing helpers, raised the share of verified results. During a first measurement, 41% of the problems were solved. The three results counted as wrong contain special functions evaluated outside their real domain (${}_2F_1$ at arguments $e^x > 1$, the incomplete Gamma function of a negative argument), where the check compares different complex branches. A run takes about ten minutes.

Result	Problems	Share
solved and verified	847	55.0%
no antiderivative found	673	43.7%
wrong result	3	0.2%
crash or time-out	7	0.5%
not verifiable	9	0.6%

Table 2: Results on the 1539 tested problems of four independent integration problem collections in Rubi's test suite.

The translation also served as a test of the library itself. It revealed two incorrect simplification rules in the standard algebra, which turned $\ln(x+y)$ into $\ln x \cdot \ln y$ and a^{b^c} into a^{bc}, two incorrect derivatives (of arcsin and arccos), and three rules that are only valid for positive bases: $(ab)^c = a^c b^c$, $(a^b)^c = a^{bc}$, which turned $\sqrt{x^2}$ into x, and $\ln(ab) = \ln a + \ln b$. They now apply only for integer exponents or a positive numerical factor. The numerical check of the harness found these errors, because Rubi's results were differentiated at negative arguments.

4 Implementation and revival

The interpreter consists of about 8800 lines of C, of which about 3900 were added in 2026 for the rule lists, subsorts, exact numbers, special and helper functions, and a grammar of about 490 lines for the scanner and parser generators flex and bison. Terms are binary trees of nodes taken from memory pools; symbols carry their sort, value and the links used by `LIFT`. The rules of an algebra are merged into a single tree, in which rules with a common left-hand-side structure share nodes and alternatives branch off where they differ; rule lists are kept in order instead. Rules are matched against every subterm, innermost first, and a single `REWR` stops after 10^5 steps, so that a looping rule set cannot hang the interpreter.

The program was written for 32-bit Unix and the Amiga and no longer worked with current tools. The port to 64-bit Linux and Windows (MSYS2) required only small changes to the C code itself. The main problems were changed semantics in the scanner generator, the release of static memory that old C libraries tolerated, and errors in the handling of declarations inside loops and included files. A regression test suite with 28 test programs now covers the core language, the library, the pattern features, exact numbers, helper and special functions, the integration rules and the original example programs; the integration tests check every result by differentiating it again. Memory leaks were removed with an instrumented build that counts allocations per source line, so that repeated calculations run in constant memory. All parser structures are now accessed through named macros instead of chains like `np->r->r->l`; for most source files the compiled code stayed byte-identical, which confirmed that this cleanup did not change the behaviour.

Performance is adequate for interactive use. Loading the standard library and computing a derivative takes well below 0.1 s on a current laptop; loading the 6713 integration rules about half a second. A scalar loop written in the language itself needs about 0.25 μs per iteration (measured for 10^7 iterations of `s = s + k*k`). We estimate this to be about two orders of magnitude slower than an equivalent compiled loop. Operations on whole vectors and matrices run in C.

5 Discussion

Strengths. The main strength of `rewrite` is that all mathematics is visible and changeable. Correcting the derivative of arcsin meant editing one line of a text file, not the interpreter. `LIFT` allows algebraic structure to be expressed once and reused; with subsorts, it also turned out

to be the right mechanism for a numeric tower, since exact fractions inherited all rules of the real numbers without any change to them. `NUM` variables together with conditions give precise control over when a rule fires. The interleaving of rewriting and numerical evaluation makes it natural to derive formulas symbolically and evaluate them numerically in the same program, for example derivatives for Newton's method.

Limitations. Commutative matching only swaps the two operands of one binary operator. Associativity is handled by normalization, so the integrand must be brought into a fixed form before the rules are tried. Complex numbers exist as a sort of the library, but not yet as part of the number tower, so rules that need the imaginary unit are not translated, and logarithms of negative arguments evaluate to NaN although their derivatives are correct. Each sort can have only one wider sort; a sort of non-negative numbers, which would make rules like $\sqrt{x^2} = x$ applicable where they are valid, would need several. Nothing checks that a rule set terminates or is confluent [1], and the order of rules matters; the safeguards against loops in rule lists are heuristics. Numeric conditions such as $m \neq -1$ do not hold for symbolic m, which is more conservative than Rubi. Error messages are minimal.

Related work. Rule-based simplification is at the core of Mathematica [17] and of Rubi [11]. General rewriting logic, with order-sorted signatures, is provided by Maude [5, 6]. More recently, equality saturation with e-graphs, as in egg [16] and Metatheory.jl [4], avoids the dependence on rule order by representing many equivalent terms at once. Proof assistants organize mathematical knowledge in algebraic hierarchies and simplification rule sets [15]. `rewrite` anticipated several of these ideas in a small and self-contained system.

Outlook. Natural next steps are complex numbers on top of the number tower and subsorts with several supersorts, for example for non-negative numbers; the translation of Rubi rules with function variables, which can be expanded into one rule per admissible function; Rubi's substitution machinery; an associative-commutative matcher for sums and products of any length; and code generation that turns symbolically derived expressions into compiled numerical code.

6 Conclusion

`rewrite` shows that a small interpreter can host a complete computer algebra library when the mathematics is expressed as rewrite rules and rule sets can be combined and lifted between algebraic structures and number sorts. The automatic translation of 90% of Rubi demonstrates that established rule-based algorithms can be imported, and it clarifies which matching features and which implicit simplifications of the host system such an import requires.

Availability

The source code, the standard library, the integration rules, the translator, the test harness and the test suite are available at `https://github.com/RainerHeintzmann/rewrite` under the MIT license [8]. The translated integration rules are derived from Rubi [13] and keep its MIT license notice.

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